

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Reduce the Order

1. Order Reduction of Linear 2nd-Order ODEs

Consider the linear 2nd-order ODE with non-constant coefficients:

$$t^2 y'' + 2ty' - 2y = 0$$

with a solution $y_1 = t$.

The following tasks outlines the method of *reduction of order* to solve for the general solution of the given ODE.

Task	Show work here
Assume that the other solution is of the form $y_2 = tv$ where v is some function of t . Determine y_2' and y_2'' . Then substitute these into the given ODE. Simplify.	
Use the variable transformation $w = v'$ to reduce the ODE into 1st-order.	
Determine the solution w of the 1st-order ODE.	
Since $w = v'$, solve for $v = \int w \, dt$. You can choose any constants to clear out fractions. If a constant is by itself choose 0.	
Write $y_2 = y_1 v$.	
Write the general solution $y = c_1 y_1 + c_2 y_2$.	

This method is useful when one solution is already known and you need to find a second solution.

2. Using a Different Trial Solution

Consider the general form of a linear 2nd-order ODE with non-constant coefficients: $t^2 y'' + aty' + by = 0$ where a and b are constants. Let $y = t^r$ be the trial solution. This will result into the characteristic equation

$$r^2 + (a - 1)r + b = 0.$$

The roots of the characteristic equation, determines the form of the general solution:

- If r yields *two distinct real roots* r_1 and r_2 , the general solution is

$$y = c_1 t^{r_1} + c_2 t^{r_2}.$$

- If r yields a *repeated real root*, the general solution is

$$y = (c_1 + c_2 \ln(t)) t^r.$$

- If r yields *complex conjugate roots* $r = \alpha \pm \beta i$, the general solution is

$$y = t^\alpha (c_1 \cos(\beta \ln(t)) + c_2 \sin(\beta \ln(t))).$$

Consider the same linear 2nd-order ODE with non-constant coefficients from Problem (1):

$$t^2 y'' + 2ty' - 2y = 0.$$

- a. Let $y = t^r$ be the trial solution where r is a constant. Use the verification process to determine the characteristic equation.

- b. Solve for the roots of the characteristic equation determined in Part (a). Then, write the general solution. Compare this solution with the solution determined in Problem (1).

- c. Now, consider this different ODE of the same form

$$t^2 y'' - ty' + 3y = 0.$$

State the characteristic equation, find its roots, and write the general solution.